INCH-POUND

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MILITARY SPECIFICATION

OVERSHOES, MEN'S AND WOMEN'S, BOOT, COMBAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers one type of slip resistant, fuel resistant, chemical protective, molded vinyl plastisol overshoe.

1.2 Classification. The overshoe shall be of the following sizes as specified (see 6.2).

Schedule of sizes

3	6	9	12
4	7	10	13
5	8	11	14

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 84GP

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SPECIFICATIONS

FEDERAL

* A-A-50083 - Bag, Plastic, Folded Garment
UU-P-553 - Paper, Wrapping, Tissue
PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-F-495 - Finish, Chemical, Black, for Copper Alloys
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-601 - Rubber: Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
* MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance-Test Methods
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-1-1371 - Overshoes, Boot, Combat

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-5014.)

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* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 638	- Tensile Properties of Plastics
D 746	- Brittleness Temperature of Plastics and Elastomers by Impact
D 1004	- Initial Tear Resistance of Plastic Film and Sheeting
D 2240	- Rubber Property - Durometer Hardness
D 3951	- Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Samples.

3.2.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variations from this specification may appear in the sample, in which case this specification shall govern.

3.2.2 Standard sample. The traction sole tread design of the finished overshoe shall match the traction sole tread design of the standard sample. The standard sample for tread design shall be one completely fabricated overshoe. The overshoe shall also match the standard sample for color and finish. The standard sample for color and finish shall be a swatch of vinyl (see 6.4).

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

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3.3.1 Vinyl. The molding compound shall consist of virgin vinyl resins, plasticizers, stabilizers, and pigments which are suitable for either slush or injection molded overshoes conforming to the requirements of this specification. Testing shall be as specified in 4.4.1.

3.3.2 Button assembly. The button assembly shall consist of a 27 line, open top brass button conforming to Universal style UB 807 or equal; and an 18 line, two prong capped fastener conforming to Universal style UB 878 or equal. The two components shall have a black chemical finish conforming to MIL-F-495. Prior to the use of the "or equal", the contractor shall furnish a sample of the button assembly with the supporting data to the contracting officer for subsequent evaluation by the responsible Military agency. The assembly of the components shall be as shown in Drawing 2-1-1371. The button assembly, after being secured to the overshoe, shall withstand a minimum pull of 35 pounds without separation of the button from fastener or the fastener pulling through the overshoe when tested as specified in 4.4.5.

3.3.3 Fastening loop. The fastening loop shall consist of a green nylon braid with an elastic core and a corrosion resistant, black, metal clip suitable for forming around the braid without fracturing. The fastening loop construction and dimensions shall be as shown in Drawing 2-1-1371. The completed fastening loop shall withstand a static load of 25 pounds for not less than 30 minutes without breaking when tested as specified in 4.4.1.

3.4 Design. The overshoe shall be a vinyl plastisol, slush or injection molded, pull-on type with a folded gusset and three buttons on each side of the gusset. Closure shall be accomplished with three fastening loops (see Drawing 2-1-1371). The design of the overshoe shall be as shown on Drawing 2-1-1371.

3.5 Molds. The molds used for slush or injection molding the overshoe shall be furnished by the contractor. The molds shall be fabricated so as to produce a finished overshoe conforming to the design shown on Drawing 2-1-1371.

3.6 Construction.

3.6.1 Molding. The overshoe shall be slush or injection molded in such a manner that the dimensions of the overshoe including the thickness conform to Drawing 2-1-1371. The molding process shall be performed in a one-feed operation so that the vinyl plastisol formulation is uniformly cured throughout the thickness of the overshoe. The cured vinyl plastisol shall have a uniform appearance throughout the thickness of the cross section of the material and shall not peel, blister, delaminate, or separate when tested as specified in 4.4.5. The overshoes shall not have any drips or runs on the inner surface which are so thick that they cause a noticeable increase in stiffness.

3.6.2 Attachment of fastening loops. Three fastening loops, one per button, shall be attached to the three buttons on the inward side of each overshoe by passing the smaller of the two formed loops over the button.

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3.6.3 Extra fastening loops. Two extra fastening loops shall be placed inside the right overshoe of each pair of overshoes before the overshoes are packaged.

3.6.4 Color and finish. The color of the overshoe shall be Olive Green 420. The outer surface of the overshoe shall be sprayed with a clear lacquer so as to produce a lusterless appearance. The overshoe shall match the standard sample (see 3.2.2 and 6.4) for color and finish. The inside of the overshoe shall be coated with a slip finish to permit easy donning and doffing. The overshoe shall not have any tacky or sticky area.

3.6.5 Marking and identification. The size and the contractor's name or identification symbol shall be embossed in raised lettering in the shank area on the outside of the overshoe. The characters shall be 1/4 to 1/2 inch in height.

* 3.6.6 Instruction sheet. A paper instruction sheet shall be placed inside each unit package of overshoes so as to be readily visible. The size of the characters shall be 1/4 inch (18 point) for the heading and 1/8 inch (10 point) for instructions. The instruction sheet shall contain the following information:

OVERSHOES, MEN'S AND WOMEN'S
BOOT, COMBAT

a. These overshoes may be worn in the event chemical protection (CB protection) is required.

b. Always wear the overshoes under your outer trousers for chemical protection.

c. Periodically inspect your chemical protective overshoes for holes or punctures. In the event they do develop holes or punctures, discard immediately and use a serviceable pair.

d. Overshoes exposed to toxicological agents shall be discarded and replaced with a serviceable pair.

e. Should your overshoes become contaminated with gasoline, oil, grease, or cleaning fluids, wipe off and air dry within two minutes.

DONNING INSTRUCTIONS

1. SELECT OVERSHOE CLOSEST TO YOUR BOOT SIZE.

2. INSERT BOOT INTO OVERSHOE. MAKE SURE THE FIT IS NEITHER TOO TIGHT NOR TOO LOOSE.

3. CLOSE OVERSHOE WITH LOOP CLOSURES AND SEE THAT OVERSHOE COMPLETELY COVERS THE FOOTWEAR AND FUNCTIONS DURING LOCOMOTION.

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* 3.6.7 Marking and identification (for Navy procurement only (see 6.2). Each overshoe shall have the following inscription marked in indelible white ink (see 6.7) on the outer leg portion of the overshoe, centered to the left of the bottom button. The size of the lettering shall be 3/8 inch in height.

FOR CHEMICAL WARFARE PROTECTION ONLY

3.6.7.1 Identification symbol. The size and the contractor's name or identification symbol shall be embossed in raised lettering in the shank area on the outside of the overshoe. The characters shall be 1/4 to 1/2 inch in height.

* 3.6.7.2 Identification sheet. The polybag of each pair of boots shall contain an identification sheet of white paper approximately 4 X 5 inches with the following information printed in black ink. The lettering size shall be no less than 1/8 inch in height (see 5.1.1.1):

NOMENCLATURE
SIZE
CONTRACT NO.
NATIONAL STOCK NO.
MANUFACTURER
DATE MANUFACTURED

The information contained on the sheet shall be positioned in the polybag so that it is visible to users.

* 3.6.7.3 Wearing instructions. The polybag of each pair of overshoes shall contain an 8-1/2 by 11 inch sheet of white paper with the following instructions printed in black ink. The lettering size shall be no less than 1/8 inch in height (see 5.1.1.1):

OVERSHOES, CHEMICAL PROTECTIVE, SIZES 3 TO 14

WEARING INSTRUCTIONS

1. THESE OVERSHOES ARE TO BE WORN OVER CONVENTIONAL FOOTWEAR IN THE EVENT CHEMICAL PROTECTION (CB PROTECTION) IS REQUIRED. THE SIZE RANGE IS 3 TO 14. SELECT A SIZE CLOSE TO THE BOOT SIZE.

2. ALWAYS WEAR THE OVERSHOES UNDER YOUR OUTER TROUSERS.

3. PERIODICALLY, INSPECT YOUR CHEMICAL PROTECTIVE FOOTWEAR OVERSHOES FOR HOLES OR PUNCTURES. IN THE EVENT THEY DO DEVELOP HOLES, ARE PUNCTURED, OR BECOME TORN, DISCARD IMMEDIATELY AND REPLACE WITH A SERVICEABLE PAIR.

4. SHOULD YOUR OVERSHOES BECOME CONTAMINATED WITH GASOLINE, OIL, GREASE OR CLEANING FLUIDS, WIPE OFF AND AIR DRY WITHIN TWO MINUTES. IF THEY CANNOT BE WIPED OFF AND DRIED WITHIN TWO MINUTES THEY SHOULD BE REPLACED IMMEDIATELY.

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DONNING INSTRUCTIONS

1. SELECT OVERSHOE CLOSEST TO YOUR BOOT SIZE.

2. INSERT BOOT INTO OVERSHOE. MAKE SURE THE FIT IS NEITHER TOO TIGHT NOR TOO LOOSE.

3. CLOSE OVERSHOE WITH LOOP CLOSURES AND SEE THAT OVERSHOE COMPLETELY COVERS THE FOOTWEAR AND FUNCTIONS DURING LOCOMOTION.

3.7 Physical characteristics.

3.7.1 Tensile strength at 100 percent elongation. The tensile strength of the overshoe at 100 percent elongation shall be greater than 250 psi but less than 500 psi when tested as specified in 4.4.5.

3.7.2 Tensile strength at break. The overshoe shall have an average tensile strength at break greater than 900 psi when tested as specified in 4.4.5. No individual test value shall be less than 750 psi.

3.7.3 Tensile strength at 100 percent elongation after immersion in Medium No. 6. The tensile strength of the overshoe at 100 percent elongation after immersion in Medium No. 6 shall not increase by more than 125 percent over the initial tensile strength at 100 percent elongation (see 3.7.1) when tested as specified in 4.4.5.

3.7.4 Elongation at break. The overshoe shall have an elongation at break greater than 300 percent when tested as specified in 4.4.5.

3.7.5 Tear strength. The overshoe shall have an average tear strength greater than 110 pounds per inch of thickness when tested as specified in 4.4.5. No individual tear strength value shall be less than 95 pounds per inch of thickness.

3.7.6 Hardness. The overshoe shall have a Shore A durometer reading of not greater than 65 when tested as specified in 4.4.5.

3.7.7 Brittleness temperature. The overshoe shall have a brittleness temperature of -40°C or lower when tested as specified in 4.4.5.

3.7.8 Leakage. The overshoe shall show no evidence of leaking when tested as specified in 4.4.5.

* 3.7.9 Mustard (HB) and GB resistance. The overshoe shall have a mustard (HB) resistance (break time in minutes) of 210 minimum and a GB resistance (break time in minutes) of 360 minimum when tested as specified in 4.4.5.

3.8 Replacement of defective components. During the manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with nondefective and properly matched components.

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3.9 Workmanship. The end item shall conform to the quality of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the visual defects specified in 4.4.3, the thickness and finished dimensional requirements specified in 4.4.4, and tested for the characteristics specified in 4.4.5.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, testing shall be performed on components and materials listed in table I. The sample unit for dimensional testing of the button assembly shall be one button and one fastener. The sample unit for testing the fastening loop shall be one completely fabricated fastening loop. All requirements are applicable to the sample unit. The lot shall be unacceptable if one or more sample units or the vinyl compound fails to meet any requirement specified. The sample size shall be as follows:

Lot size	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE I. Component tests

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	No. of determinations per sample unit
Vinyl compound (batch)	Material identification	3.3.1	1/	-
Button assembly (one button and fastener)	Material identification	3.3.2	1/	-
	Dimension	3.3.2	gage 2/	1
Fastening loop (one fastener)	Material identification	3.3.3	1/	-
	Breaking strength	3.3.3	4.5.1	1
	Dimension	3.3.3	gage 2/	1

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Results shall be reported to the nearest 0.001 inch or 1/32 inch as applicable.

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4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

* 4.4.2.1 End item critical defect examination. Every end item (100%) prior to the visual and dimensional examinations in 4.4.3 and 4.4.4, shall be examined for the critical defects listed in table II. Any end item found to contain one or more critical defects shall be rejected.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of overshoes. The sample unit shall be one completely fabricated overshoe and the selection shall be done by pairs. The inspection level shall be II (see 6.5). For the pairing examination, the pair shall be examined together. Defects of pairing shall be classified as a single defect. The finding of one or more critical defects shall be cause for rejection of the lot.

NOTE: Appearance defects shall be scored only when the condition is plainly visible at a distance of 3 feet or more.

TABLE II. End item visual defects

Defects designated by an asterisk (*) shall be scored as a major when seriously affecting serviceability or appearance, and minor when affecting serviceability or appearance but not seriously.

Examine	Defect	Classification		
		Critical	Major	Minor
Pairing	Not properly mated, i.e., not right and left of same size		101	
	Variation in color between pair			201
	Variation in height over 1/4 inch			202
Cleanness	Any spot or stain seriously affecting appearance			203
	Any embedded foreign matter not readily removed without leaving a thin spot	1		

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Design	Any characteristic not in accordance with specified requirements (i.e., overshoe not slush or injection molded, not pull-on type with folding gusset, and does not have three buttons on each side)		102 *	204
Color and finish (applicable to all components)	Color not uniform or not as specified			205
	Inside of overshoe not coated with a slip finish			206
	Outside finish glossy		103	
	Finish tacky or sticky		104	
Fastening loop	Braid cut or frayed, fastener (except extra fastening loop) missing or not attached as specified	2		
	Clip missing, mispositioned, or not securely attached	3		
	Extra fastening loops not in overshoe			207
Vinyl	Any hole, cut, tear, rupture, or crack	4		
	Any blister, weak or thin spot, or bubble	5		
	Any repair or patch	6		
	Drip or run on the inner surface which causes a noticeable increase in stiffness			208
Construction and workmanship	Any part of button malformed, broken, missing, or otherwise defective, i.e., too loose	7		
	Any button clinched too tightly		105	
	Any component or assembly omitted or misplaced; any operation omitted or not properly performed (unless otherwise classified herein)		106 *	209

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Marking, (identification and size)	Missing, incorrect, illegible, or misplaced			210
	Chemical warfare inscription not indelible; i.e., easily removed when rubbed (for Navy procurements only)	8		

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the thickness and finished dimensional requirements of Drawing 2-1-1371. The thickness shall be measured using a dial micrometer with a throat at 12 inches deep. The anvils of the micrometer shall be 3/8 inch diameter. Any dimension or thickness that is not within the established tolerance shall be classified as a defect. The lot size shall be expressed in units of overshoes. The sample unit shall be one overshoe and the selection shall be by pairs. The inspection level shall be S-3 (see 6.5).

* 4.4.5 End item testing. The end items shall be tested for the characteristics listed in table III. The lot size shall be expressed in units of overshoes. All requirements shall be applicable to the sample unit. All test reports shall contain the individual values used to express the final results. The Government shall perform acceptance tests for mustard (HB) resistance and GB resistance (see 6.8). Test specimens of proper size to accomodate the test apparatus for the mustard and GB resistance tests shall be taken from the upper side panel within 6 inches of the top of the overshoe. The sample unit for these tests shall be two pairs of overshoes and the inspection level shall be S-2. These tests are considered critical and any failure shall be cause for rejection of the lot represented by the sample. For all other tests, the sample unit shall be one pair of overshoes. The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified:

Lot size	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III. End item tests

Characteristic	Requirement paragraph	Test method	No. of determinations per sample unit	Results reported as
Pull test (button assembly)	3.3.2	4.5.2	3	Pass or fail
Uniform molding and curing test	3.6.1	4.5.3	1	Pass or fail
Tensile strength: At 100 percent elongation	3.7.1	ASTM D 638 1/	3	To nearest 10 psi
At break	3.7.2	ASTM D 638 1/	3	To nearest 10 psi
At 100 percent elongation after immersion in Medium No. 6	3.7.3	4.5.4	3	To nearest 10 psi
Elongation at break	3.7.4	ASTM D 638 1/	3	To nearest 1 percent
Tear strength	3.7.5	ASTM D 1004 1/	3	To nearest 1 pound
Hardness	3.7.6	ASTM D 2240	3	To nearest whole number
Brittleness temperature	3.7.7	ASTM D 746	1	To nearest 1°C
Leakage test	3.7.8	4.5.5	1	Pass or fail
Mustard resistance (HB)	3.7.9	T204 or T209 2/	2	To nearest 1 minute
GB resistance	3.7.9	T206 or T208 2/	2	To nearest 1 minute

1/ The speed of the machine shall be 12 inches per minute.

2/ Refers to test methods in MIL-STD-282.

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4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 (see 6.5).

Examine	Defect
Marking (exterior and unit packs)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

* 4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 (see 6.5).

Examine	Defect
Finished dimensions	Length, width, or height exceed specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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4.5 Methods of inspection.

4.5.1 Breaking strength test of fastening loop. One loop of the fastening loop shall be placed over a fixed suspension hook. A second hook suitable for attaching weights shall be passed through the other loop and weights applied in increments not exceeding 10 pounds until a total load of 25 pounds has been applied. The weights shall hang freely exerting a 25-pound load on the fastening loop for a period of 30 minutes. Any loop that breaks during the 30-minute period shall be a test failure.

4.5.2 Pull test. The vinyl specimen shall be 4 inches long and 1-1/2 inches wide and shall be cut from the overshoe so that the middle of the button is in the center of the specimen. The specimen shall be mounted in the button pull test fixture as shown on Drawing 2-1-1371 and the test conducted using a tensile testing machine operating at a speed of 12 inches per minute. A test failure shall be reported if either the button and fastener separate or if the fastener pulls through the vinyl upper at a pull of 35 pounds or less.

4.5.3 Uniform molding and curing test. The sole from an overshoe from each sample unit shall be cut with a sharp knife along the center line from the toe to the heel. Evidence of any blister, peeling, delamination, or separation or material shall be a test failure.

4.5.4 Tensile strength test at 100 percent elongation after immersion in Medium No. 6. The dumbbell shaped specimen shall be as specified in ASTM D 638 except that the specimen shall be 4-3/8 inches long. The specimen shall be immersed in fuel conforming to Medium No. 6 as specified in Method 6001 of FED-STD-601. The temperature of the medium shall be 23 ± 2°C and the time of immersion shall be 16 hours. At the end of the required immersion time, the specimen shall be removed from the medium, dried with a soft absorbent cloth and allowed to air dry for 4 hours. The specimen shall then be tested for tensile strength at 100 percent elongation in accordance with ASTM D 638 except that the speed of the machine shall be 12 inches per minute. The thickness of the specimen shall be determined after the specimen has been air dried for 4 hours.

4.5.5 Leakage test. The specimen shall be an overshoe with all fastening loops in place so as to keep the gusset closed. The overshoe shall be filled with water at 23 ± 2°C to within 1/2 to 1/4 inch of the middle button assembly. After 15 minutes, the outside of the water filled overshoe shall be examined. Any evidence of leakage shall be a test failure.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

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* 5.1.1 Level A preservation. Each overshoe shall have a non-bending grade paperboard stay inserted the full height of the uppers. The paperboard stay shall be 0.032 inch minimum thickness and shall measure approximately 5 inches by 11-1/2 inches. One pair of overshoes shall be unit packed in a plastic bag conforming to A-A-50083 with a minimum material thickness of 3 mil. An identification sheet and donning instructions sheet shall be added to each bag (see 3.6.6). Each overshoe shall be separated from the other by tissue paper conforming to type I, class 1 of UU-P-553.

* 5.1.1.1 Level A preservation (for Navy procurement only). Button overshoes prior to folding. Each overshoe shall have the top portion folded inward under the bottom portion of the overshoe. After folding, each pair shall be placed bottom to bottom and wrapped with two rubber bands of a suitable length and width for the purpose intended. Each band shall be positioned approximately 3 inches from each end of the bundle. The bundle shall be approximately 10 inches in length by 13 inches in width. Each pair shall be placed in a polyethylene bag of 4 mil (.004 inch) thickness, (± 25 percent tolerance) approximately 9 inches X 10-1/2 inches with a 2 inch gusset. An identification sheet and wearing instructions sheet shall be added to each bag (see 3.6.7.2 and 3.6.7.3). Each poly bag shall have the excess air removed and shall then be heat sealed.

5.1.2 Commercial preservation. Overshoes shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Six pairs of overshoes of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to grade V2s, style RSC-L of PPP-B-636. Inside dimensions of the box shall be approximately 24 inches in length, 13 inches in width, and 13 inches in depth for sizes 3 through 8, and 28 inches in length, 14 inches in width, and 15 inches in depth for sizes 9 through 14. Each box shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.2.2 Level B packing. Six pairs of overshoes of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to type CF (variety SW) or SF, class domestic, grade 275, style RSC-L of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Inside dimensions of the box shall be approximately 24 inches in length, 13 inches in width, and 13 inches in depth for sizes 3 through 8, and 28 inches in length, 14 inches in width, and 15 inches in depth for sizes 9 through 14. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

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* 5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636,except that the inspection shall be in accordance with 4.4.6.

5.2.2.2 Level B packing (for Navy procurement only). One polybag liner containing twelve pairs of overshoes of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container (regular slotted carton) conforming to type CF, class domestic, variety DW, grade 275, size 3A of PPP-B-636. Inside dimensions of each shipping container shall approximate 19-1/2 inches in length, 15 inches in width, and 15-5/8 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Each container shall be closed in accordance with method III and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial packing. Overshoes, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), overshoes packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be 3 for sizes 3 through 8 and number 6 for sizes 9 through 14.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended use. The overshoe is intended to be worn over the MIL-5 combat boot by both men and women in the Military. The overshoes will provide moisture protection and may be used for chemical protection.

* 6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).

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c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Marking and identification for Navy procurements (see 3.6.7).
f. Levels of preservation and packing (see 5.1 and 5.2).
g. Preservation and packing for Navy procurements (see 5.1.1.1 and 5.2.2.2).
h. Type and class of unit load required (see 5.2.1).
i. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
j. When palletization is required (see 5.3).
k. Acceptance criteria required (see 6.5).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Acceptance criteria. The acceptance criteria below are recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with MIL-STD-105.

6.5.1 For end item visual examination. An acceptable quality level (AQL), expressed in terms of defects per hundred units, of 2.5 for major defects and 6.5 for total (major and minor combined) defects is recommended.

6.5.2 For end item dimensional examination. An AQL, expressed in terms of defects per hundred units, of 4.0 is recommended.

6.5.3 For packaging examination. An AQL, expressed in terms of defects per hundred units, of 2.5 is recommended.

6.5.4 For palletization examination. An AQL, expressed in terms of defects per hundred units, of 6.5 is recommended.

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* 6.6 Subject term (key word) listing.

Chemical warfare protection
Footwear
Feet, protection
Moisture protection
Rubber
Vinyl

* 6.7 Marking (for Navy procurement only). An indelible white ink for marking the overshoes, (see 3.6.7) may be obtained from Raffi & Swanson, Co. Inc., 100 Eames St., Wilmington, Ma. 01887 (style number 9093-12W).

* 6.8 Government acceptance tests (see 4.4.5). Government acceptance tests shall be performed at the US Army Armament, Munitions and Chemical Command, (AMCCOM) (Edgewood Area), Chemical Test Branch, Bldg. E5100, Aberdeen Proving Ground, MD 21010-5423.

6.9 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - GL
Navy - NU
Air Force - 99

Preparing activity:
Army - GL

(Project 8430-0381)

Review activities:
Army - MD
Air Force - 11, 82
DLA - CT

User activity:
Navy - MC
Air Force - 45

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-43995C	Overshoes, Men's & Women's, Boot, Combat

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
3b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	7b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
7c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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